

B.Tech II Year I Semester (R20) Supplementary Examinations November 2024

COMPLEX VARIABLES & TRANSFORMS

(Common to EEE & ECE)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- (a) If $f(z) = \sin z$ then prove that $f(z)$ is a regular function and find $f'(z)$. 2M
- (b) Show that the function $u = x^2 - y^2 - 2xy - 2x + 3y$ is harmonic 2M
- (c) Explain the difference between a regular singular point and an irregular singular point of a complex function. Provide an example of each. 2M
- (d) Find the Laurent series expansion of the function $f(z) = z/(z^2 - 1)$ about the point $z = 0$. 2M
- (e) Find $L[e^{2t} \cos 3t]$ 2M
- (f) Find the $L[t^2 u(t-3)]$ 2M
- (g) Obtain the Fourier coefficient a_n for $f(x) = x^2$ in $-\pi \leq x \leq \pi$ 2M
- (h) Check the even and odd nature of the function $f(x) = x - x^2$ in $(-\pi, \pi)$ 2M
- (i) Find the Fourier sine transforms of $f(x) = \begin{cases} x, & 0 < x < 2 \\ 0, & \text{else where} \end{cases}$ 2M
- (j) Find the Z-Transform of $\sin(3n + 5)$. 2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Show that $u = \left(r + \frac{1}{r}\right) \cos \theta$ is harmonic. Find its harmonic conjugate and also the corresponding analytic function. 10M

OR

- 3 If $\phi(x, y)$ is a differentiable function and $f(z) = u(x, y) + iv(x, y)$ is a regular function, show that 10M

$$\left(\frac{\partial \phi}{\partial x}\right)^2 + \left(\frac{\partial \phi}{\partial y}\right)^2 = \left\{ \left(\frac{\partial \phi}{\partial u}\right)^2 + \left(\frac{\partial \phi}{\partial v}\right)^2 \right\} |f'(z)|^2$$

- 4 Evaluate $\int_C \frac{e^{2z}}{(z+1)^2(z-2)} dz$ where $C: |z| = 3$. 10M

OR

- 5 Expand $f(z) = \frac{z+1}{(z+2)(z+3)}$ in a Laurent's series valid for 10M

a) $|z| > 3$

b) $2 < |z| < 3$

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- 6 If $f(t)$ is a periodic function of period $T > 0$, then prove that 10M

$$L(f(t)) = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt.$$

OR

- 7 Solve the following initial value problem by using Laplace transform method 10M
 $y'' + 4y' + 4y = e^{-t}, y(0) = 0, y'(0) = 0.$

- 8 Obtain the Fourier Series for the function $f(x) = 2x - x^2$ in $0 \leq x \leq 3$. 10M

OR

- 9 Obtain the sine half range the Fourier Series of $f(x) = x^2$ in $0 < x < \pi$. 10M

- 10 Find the Fourier transform of the function $f(x) = \begin{cases} 1-|x|, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$ 10M

Hence deduce that
$$\int_0^{\infty} \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2}$$

OR

- 11 Obtain the Z-Transform of $\cos n\theta$ and $\sin n\theta$. Hence deduce the Z-Transform of (i) $k^n \cos n\theta$ 10M
(ii) $e^{-an} \sin n\theta$.

B.Tech II Year I Semester (R20) Supplementary Examinations April/May 2024

COMPLEX VARIABLES & TRANSFORMS

(Common to EEE & ECE)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- (a) Show that $f(z) = z^3$ is analytic for all z . 2M
 - (b) Prove that $u = x^2 - y^2 - 2xy - 2x + 3y$ is harmonic. 2M
 - (c) State (i) Cauchy integral theorem, (ii) Liouville's theorem. 2M
 - (d) Determine the poles of the function $\frac{z}{\cos z}$. 2M
 - (e) Find: $L\{t^{3/2}\}$. 2M
 - (f) If $f(t)$ is a periodic function then find $L\{f(t)\}$. 2M
 - (g) State Dirichlet's conditions. 2M
 - (h) Define Odd and Even function with an example each. 2M
 - (i) State Fourier integral theorem. 2M
 - (j) If $Z[f(n)] = F(z)$ then find $Z[a^{-n}f(n)]$. 2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Find the analytic function whose imaginary part is $e^x(x \sin y + y \cos y)$. 10M
- OR**
- 3 Find the bilinear transformation which maps the points $z = 1, i, -1$ onto the points $w = i, 0, -i$. Also find the image of $|z| < 1$. 10M
- 4 (a) Using Cauchy's integral formula, evaluate $\int_c \frac{z}{(z-1)(z-2)^2} dz$. 5M
- (b) Find the residue of $\frac{ze^{zt}}{(z-3)^2}$ at its poles. 5M
- OR**
- 5 Using the method of contour integration, Prove that $\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{a+b}$ $a > 0, b > 0, a \neq b$. 10M
- 6 (a) Find $L\{t \sin at\}$. 5M
- (b) Find $L^{-1}\left\{\frac{s^2}{(s+1)(s+2)(s+3)}\right\}$. 5M
- OR**
- 7 (a) Using convolution theorem find $L^{-1}\left\{\frac{1}{(s+1)(s^2+4)}\right\}$. 5M
- (b) Using Laplace transform, solve $(D^2 + 4D + 5)y = 5$, given that $y(0) = 0, y'(0) = 0$. 5M
- 8 Find a Fourier expansion for $f(x) = x + x^2, -\pi \leq x \leq \pi$ hence find $\sum_{n=1}^{\infty} \frac{1}{n^2}$. 10M
- OR**
- 9 Find a Fourier sine series expansion of $f(x) = x(\pi - x), 0 < x < \pi$. Hence Find $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \dots$. 10M
- 10 (a) Find the Fourier transform of $\frac{1}{x}$. 5M
- (b) Find the Fourier Sine transform of e^{-ax} . 5M
- OR**
- 11 (a) Find $Z(2.3^n + 5n)$ and deduce $Z[2.3^{n+3} + 5(n+3)]$ using shifting theorem. 5M
- (b) Find the inverse Z-transform of $\frac{Z}{(Z-1)(Z^2+1)}$. 5M

B.Tech II Year I Semester (R20) Supplementary Examinations August/September 2023

COMPLEX VARIABLES & TRANSFORMS

(Common to EEE & ECE)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- (a) Describe the harmonic function with suitable example. 2M
- (b) Discuss the continuity of a complex function. 2M
- (c) State the Taylor's series for a function $f(z)$. 2M
- (d) Obtain the residue of $f(z) = \frac{z-3}{z(z^2+1)}$ at a simple pole $z=0$. 2M
- (e) Compute the Laplace transform of $f(t) = e^{3t} + \sin 5t$. 2M
- (f) Obtain the inverse Laplace transform of $F(s) = \frac{1}{s^2 - a^2}$. 2M
- (g) Write Dirichlet Conditions for the existence of Fourier series. 2M
- (h) Write Fourier series for Even and Odd Numbers. 2M
- (i) Define Z-transform and discuss the linear property. 2M
- (j) State Fourier integral theorem of $f(x)$. 2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Suppose $w = \phi + i\psi$ represents the complex potential function for an electric field and $\psi = x^2 - y^2 + \frac{x}{x^2 - y^2}$. Determine the function ϕ . 10M

OR

- 3 If $f(z)$ is an analytic function with constant modulus then show that $f(z)$ is constant. 10M

- 4 Evaluate $\frac{z-3}{z^2+2z+5} dz$ where 'c' is $|z+1-i| = 2$ using Cauchy's integral formula. 10M

OR

- 5 Obtain the Taylors expansion of; 10M

(i) $f(z) = \frac{1}{(z+1)^2}$ about the point $z = -i$.

(ii) $f(z) = \frac{2z^3+1}{z^2+z}$ about the point $z = i$.

Contd. In Page 2

- 6 Find the Laplace transform of $f(t) = \frac{\cos at - \cos bt}{t} + t \sin at$. 10M

OR

- 7 Solve by the method of transforms, the equation; 10M
 $y''' + 2y'' - y' - 2y = 0$, $y(0) = y'(0) = 0$ and $y''(0) = 6$.

- 8 Find the Fourier series for the function: 10M

$$f(x) = -\pi, \quad -\pi < x < 0;$$

$$= x, \quad 0 < x < \pi.$$

OR

- 9 Find the Fourier series expansion of $f(x) = -x^2$, $-\pi \leq x \leq \pi$. Deduce the series 10M
 $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}.$

- 10 Find the Fourier transform $f(x)$ given by 10M

$$f(x) = \begin{cases} 1 - x^2, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases} \quad \text{Hence evaluate } \int_0^{\infty} \frac{x \cos x - \sin x}{x^3} \cos \frac{x}{2} dx.$$

OR

- 11 Determine the inverse Z-transform of $\frac{2z}{(z-1)(z^2+1)}$. 10M
